Failure Analysis Report

FAIL-0604 — Connectivity Failure

1. Failure Summary

Failure ID	FAIL-0604
Failure Type	Connectivity Failure
Severity	Major
Component	Search Indexer
Application	APP-008 — VoiceAssistant
Build Version	4.1.5
Timestamp	2024-11-17 19:46:11
Environment	ENV-006 — Cloud Test Env
OS Version	Ubuntu 22.04
Network	Ethernet
Resolution	Wont Fix

2. Test Case Reference

Test Case ID	TC-0562
Title	Ensure that data synchronization does not exceed the CPU ceiling (80%) threshold...
Execution ID	EXEC-03084
Test Run	RUN-011 — Regression Cycle 11 — Build 6.3.9
Requirement	REQ-0062 — Security — Background Sync

3. Expected vs Observed Behaviour

Expected Behaviour	The system should complete media streaming within the defined SLA.
Observed Behaviour	The system exhibited connectivity failure when the input data contains Unicode special characters.

4. Error Details

Error Message:

```
BluetoothException: device DEVICE-577 disconnected during pairing handshake.
```

Stack Trace / Log Excerpt:

```
com.samsung.health.SyncEngine.execute(line 484)
```

Log Reference	LOG-27143

5. Executed Steps at Time of Failure

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1	Verify that the UI displays 'Connection successful'.	A confirmation notification appears on the paired device.
2	Capture a screenshot and attach as evidence.	Server returns HTTP 200 with a valid JSON payload.
3	Observe the status indicator for up to 23 seconds.	A confirmation notification appears on the paired device.
4	Check server logs for event code 'EVT-9789'.	Server returns HTTP 200 with a valid JSON payload.
5	Verify that the UI displays 'Error: retry'.	No crash or ANR is observed.

6. Root Cause Analysis

Buffer overflow in the media codec when processing H.265 streams above 4K resolution.

8. Historical Similar Failures

HIST-0100	FAIL-0101	Connectivity Failure	5.1.6	2023-03-16	2	Wont Fix
HIST-0346	FAIL-0347	Connectivity Failure	3.4.4	2023-10-25	8	Resolved
HIST-0603	FAIL-0604	Connectivity Failure	6.4.4	2021-06-16	14	Resolved

9. Synthetic Data Notice

This document is entirely synthetic and intended for AI/RAG research only.

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